Rana Nameer Hussain Khan

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Education

Virginia Tech *Jan 2024 – Present*
PhD in Computer Science
Relevant Coursework: Programming Languages, Machine Learning, Fundamentals of Information Security, Statistics

National University of Sciences and Technology (NUST) *Sep 2019 – June 2023*
Bachelor's degree in Software Engineering
Relevant Coursework: Data Structures and Algorithms, Operating Systems, Computer Networks, Computer Architecture, Deep Learning, Data Mining, Software Design & Architecture

Experience

Software Engineer Intern - NuAxis Innovations *Jun 2025 – Present*

- Led the development of NuPermits, an all-in-one permitting platform for both federal and state permits.
- Worked on the development of NuAI for CX, specifically for automatic content remediation
- Contributed to proposal and bid preparation for a federal contract, aligning technical solutions with client requirements.
- Collaborated with cross-functional teams to deliver both technical implementations and strategic documentation.

Teaching Assistant – Virginia Tech *Jan 2024 – May 2025*

- Assisted in CS-1114 (Intro to Software Design) and CS-3214 (Computer Systems).
- Led 2 lab sessions every week, conducted office hours, and graded coursework.

Machine Learning Intern – UBM Technologies *Jun 2022 – Aug 2022*

- Applied ML techniques from academic research for the detection of the behavior of livestock.
- Designed algorithms for estrus and activity detection using accelerometer data.
- Used k-means clustering for unsupervised heat detection in cattle.

Publications

MACOG: Multi-Agent Code Orchestrated Generation for Reliable Infrastructure-as-Code

- Proposed and built a multi-agent LLM system for Infrastructure-as-Code(IaC) generation for reliable cloud infrastructure.
- Achieved state-of-the-art performance on IaC-Eval with reproducible tooling and open-sourced implementation.
- `https://arxiv.org/abs/2510.03902`

ProvGen: Provenance-Aware Semantics for LLM Code Generation *Under review*

- Introduced a provence-aware operational semantics for LLM-based program synthesis, enabling checkable reproducibility and audibility of generated code, and attaches structured provenance capsules to generated artifacts.

Projects

ProfitSignals – Final Year Project

- Designed and developed a cryptocurrency copy-trading application to help beginners follow professional traders.
- Implemented user authentication, real-time price tracking, and trade replication features.
- Focused on usability and responsiveness, enabling smooth performance on both web and mobile platforms.

Ask Price – Web Scraper Tool

- Developed a Python-based tool to extract product data from e-commerce websites using web scraping libraries.
- Automated parsing and formatting of product names, prices, and descriptions from multiple sources.
- Created a searchable interface to compare results and support informed purchasing decisions.

Emojify – Real-Time Emotion Detection Application

- Built and trained a convolutional neural network (CNN) in Keras to classify facial emotions using grayscale images from the FER-2013 dataset.
- Engineered a real-time pipeline using OpenCV and Tkinter to detect faces in video frames and overlay detected emotions with corresponding emojis.
- Optimized model performance with dropout regularization, data augmentation, and the Adam optimizer.

Technical Skills

Languages: Python, C/C++, Java, JavaScript, C#, SQL, MongoDB
Frameworks/Tools: FlaskAPI, React, Node.js, .NET, Git, Linux/Unix, AWS EC2, Docker, Kubernetes - EKS
Concepts: TDD, Agile Development, MVC

Awards

- 2nd place – SEECS Open House (FYP)
- Final Year Project ranked top 3 across cohort; nominated for Rector's Gold Medal